

ECON 154: The Global Economy: Trade, Finance, and Policy

Lecture 3 Notes

Markets and Prices I: Demand, Willingness to Pay, and Consumer Behavior

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Purpose of the Lecture

The previous lectures introduced scarcity, opportunity cost, incentives, marginal reasoning, evidence, and economic models. We now use those tools to study one side of a market: consumers.

This lecture focuses on demand. We will ask how consumers choose among alternatives, why a higher price normally reduces the quantity consumers wish to purchase, how individual choices combine to form market demand, and why demand can change even when the good's own price does not.

The goal is not merely to draw a downward-sloping line. The goal is to understand the choices represented by that line.

Learning Objectives

By the end of the lecture, students should be able to:

- Distinguish demand from quantity demanded.
- Explain the law of demand using opportunity cost, substitution, and purchasing power.
- Construct a demand schedule and demand curve.
- Interpret a consumer's willingness to pay.

- Explain how individual demand curves combine into market demand.
- Distinguish a movement along a demand curve from a shift of the curve.
- Identify the major factors that shift demand.
- Use simple evidence about consumer spending without confusing expenditure with quantity.

1 What Is a Market?

A market is any setting in which buyers and sellers interact. It does not need to be a physical place. A farmers' market is a market, but so are an online ticket platform, a labor market, a foreign-exchange market, and a market for apartment rentals.

Markets coordinate decisions made by many people who usually do not know one another. A buyer does not need to understand every detail of a firm's production process. A seller does not need to know everything about a buyer's preferences. Prices help communicate information between them.

This lecture examines buyers. The next lecture will examine sellers and combine the two sides of the market.

Economic Intuition

A demand curve is not simply a line on a graph. It is a compact summary of how many units consumers are willing and able to purchase at different possible prices.

2 Demand and Quantity Demanded

OpenStax defines demand as the amount of a good or service consumers are willing and able to purchase at each possible price (OpenStax, 2022). Both words matter:

- **Willing** means the consumer prefers buying the good to the available alternatives at that price.
- **Able** means the consumer has sufficient purchasing power to make the purchase.

Definition

Demand describes the complete relationship between price and the quantities consumers are willing and able to buy.

Quantity demanded is the specific quantity consumers wish to buy at one particular price.

Suppose a student would buy one concert ticket if the price were \$120, two tickets if the price were \$80, and three tickets if the price were \$40. Each price is associated with a different quantity demanded. The full set of these possible choices is the student's demand.

Common Mistake

A change in the good's own price does not normally "change demand." It changes **quantity demanded** and produces a movement along the existing demand curve. Demand changes only when some other factor shifts the entire relationship.

3 The Law of Demand

The law of demand states that, holding other relevant factors constant, a higher price generally leads to a lower quantity demanded, while a lower price generally leads to a higher quantity demanded (OpenStax, 2022).

$$P \uparrow \implies Q_d \downarrow, \quad P \downarrow \implies Q_d \uparrow.$$

The phrase "holding other relevant factors constant" is essential. If income, tastes, expectations, or the prices of related goods change at the same time, we may observe a different outcome.

3.1 Why Does Quantity Demanded Fall When Price Rises?

1. The opportunity cost of purchasing rises

When the price of a product increases, buying it requires giving up more of other goods. A \$7 coffee uses purchasing power that could have been used for lunch, transportation, or savings. The opportunity cost of that coffee is higher than when it cost \$4.

2. Consumers substitute toward alternatives

If one product becomes more expensive while similar products do not, consumers have an incentive to switch. A rise in the price of restaurant meals may encourage more cooking at home. A rise in the price of one streaming service may encourage consumers to cancel it or use a substitute.

3. Purchasing power falls

A higher price makes the consumer's income less capable of purchasing the same bundle of goods. Even without a change in nominal income, the consumer is effectively poorer with respect to that product.

Economic Intuition

The law of demand does not require consumers to perform formal calculations. It only requires that higher prices make the purchase less attractive relative to available alternatives.

4 Demand Schedules and Demand Curves

A demand schedule presents the relationship between price and quantity demanded in a table.

Price	Quantity Demanded
\$10	1
\$8	2
\$6	3
\$4	5
\$2	8

A demand curve presents the same information graphically. Economists traditionally place price on the vertical axis and quantity on the horizontal axis.

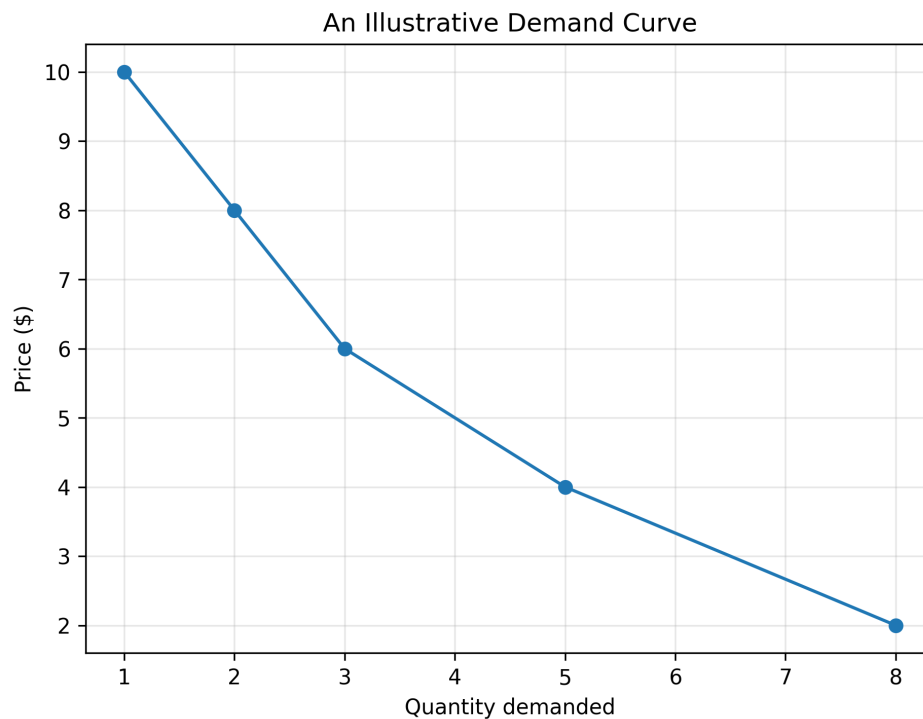


Figure 1: An illustrative demand curve constructed from a demand schedule.

The curve slopes downward because lower prices are associated with higher quantities demanded. This does not mean every consumer buys at every price. At a high price, only consumers with high willingness to pay may participate. As the price falls, additional consumers enter the market, and existing consumers may purchase more units.

5 Willingness to Pay

A consumer's willingness to pay is the maximum amount the consumer would pay for a unit rather than go without it. Willingness to pay reflects the value the consumer places on the purchase at the time of the decision. It can differ across consumers because people have different preferences, incomes, alternatives, information, and circumstances.

Consider six potential buyers of a ticket:

Buyer	Maximum Willingness to Pay
Alicia	\$18
Ben	\$15
Carla	\$12
Dev	\$9
Elena	\$7
Farah	\$4

If the market price is \$13, Alicia and Ben buy, while the other consumers do not. If the price falls to \$8, Alicia, Ben, Carla, and Dev buy. The downward-sloping demand relationship appears because progressively lower prices attract consumers with lower willingness to pay.

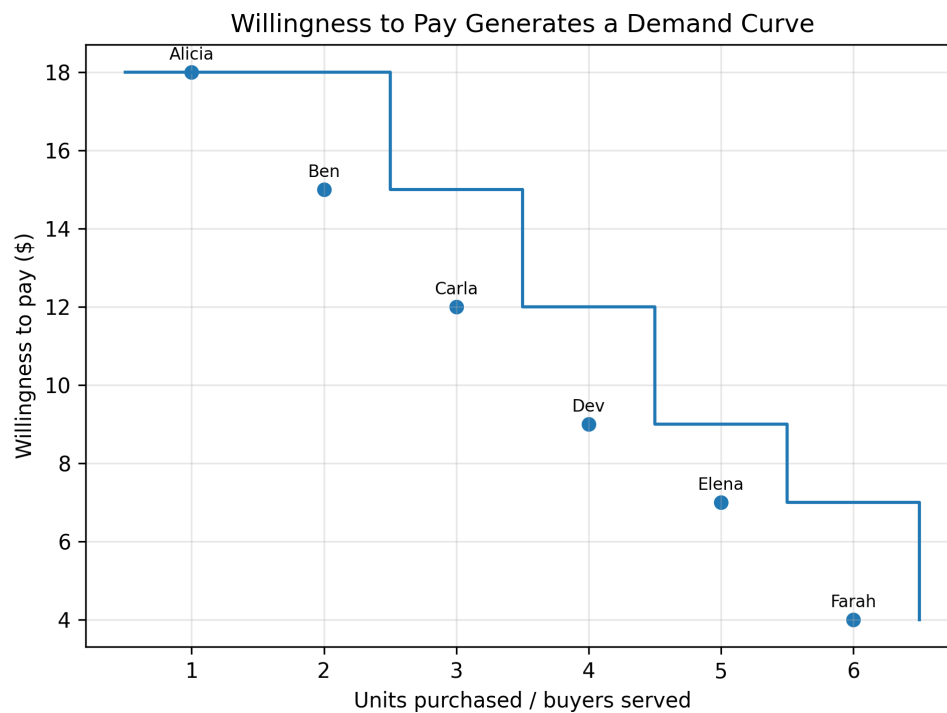


Figure 2: The willingness to pay of potential buyers generates a step-shaped demand curve.

5.1 Willingness to Pay Is Context-Dependent

A person's willingness to pay for the same product may change across situations. A bottle of water may have low value at home and much higher value during a long hike. Coffee may be worth more to a traveler facing a long airport delay than to the same person sitting near several cafés downtown. This does not mean preferences are irrational. It means value depends on circumstances and alternatives.

6 Preview: Why Voluntary Exchange Benefits Consumers

When a consumer pays less than the maximum amount they were willing to pay, the buyer receives a net benefit from the purchase.

Definition

Some consumers value a product more than the price they pay. Whenever this happens, purchasing the product leaves the consumer better off than not purchasing it.

Economists refer to this additional benefit as **consumer surplus**. In this lecture we introduce the intuition behind the concept. In Lecture 6 we will measure consumer surplus formally after explaining how supply and demand determine the market price.

Suppose Alicia would have been willing to pay \$18 for a concert ticket but purchases it for \$10. She benefits because the ticket is worth more to her than the amount she gives up to obtain it. The difference reflects part of the gain she receives from voluntary exchange.

Across an entire market, the area below the demand curve and above a given price illustrates the additional value consumers receive beyond what they pay.

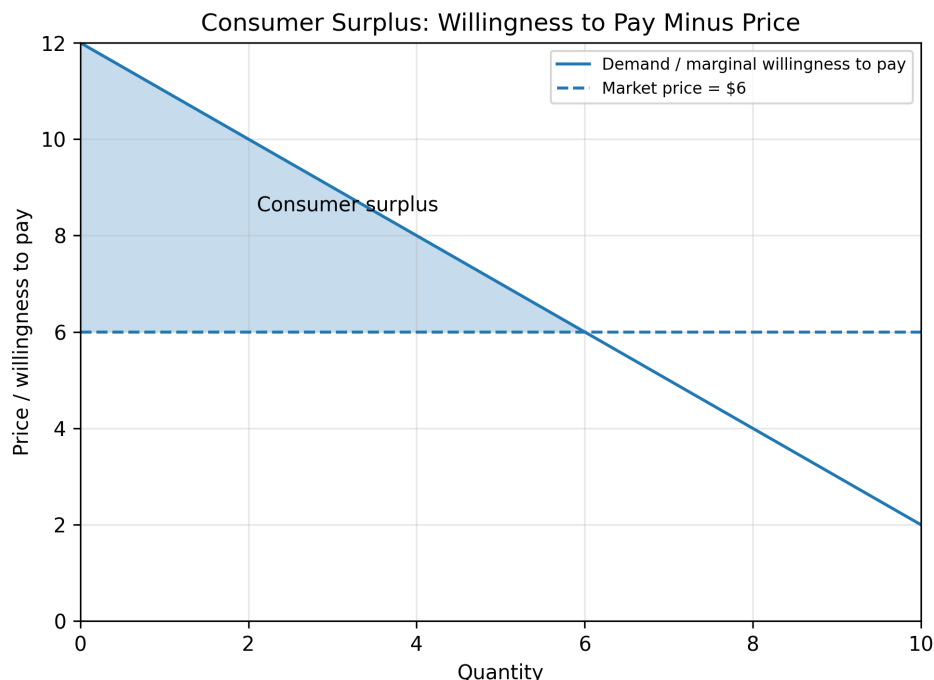


Figure 3: The shaded area illustrates the additional benefit consumers receive when they pay less than their maximum willingness to pay. We will measure this area formally in Lecture 6 after introducing market equilibrium.

At this point we have focused exclusively on consumers and treated the market price as given. A natural question remains: **where does that price come from?** Answering that question requires understanding the decisions made by firms. In the next lecture we develop the producer's side of the market. Once buyers and sellers are combined, we will return to consumer surplus and measure it using the market equilibrium.

7 Budget Constraints and Consumer Choice

Consumers may want many products, but income limits what they can purchase. A budget constraint shows all combinations of goods a consumer can afford given income and prices.

Suppose a student has \$30 to divide between coffees costing \$5 and meals costing \$10. The budget constraint is

$$5C + 10M \leq 30,$$

where C is the number of coffees and M is the number of meals.

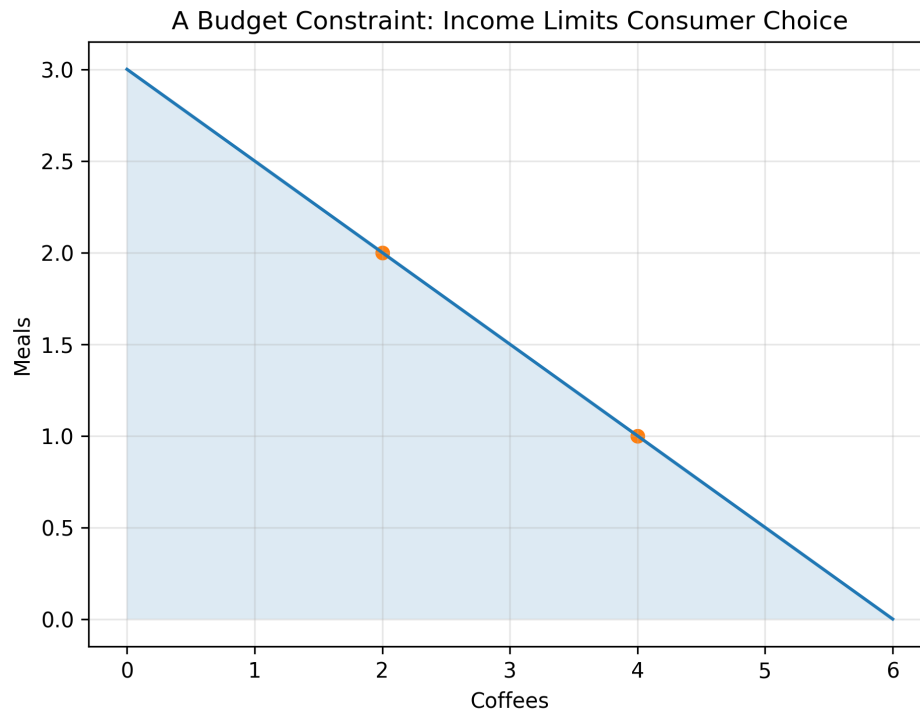


Figure 4: A budget constraint shows the affordable combinations of two goods.

A higher coffee price rotates the budget line inward along the coffee axis. The consumer can no longer afford as many coffees with the same income. This is one reason higher prices reduce quantity demanded. A higher income expands the set of affordable choices. For many goods, this increases demand. For some goods, however, consumers may switch toward higher-quality alternatives as income rises.

8 Individual Demand and Market Demand

An individual demand curve describes one consumer. Market demand describes the total quantity all consumers are willing and able to purchase at each price. To construct market demand, we add individual quantities at the same price. This is called a horizontal sum.

Price	Consumer A	Consumer B	Consumer C	Market Quantity
\$10	0	1	0	1
\$8	1	1	0	2
\$6	2	2	1	5
\$4	4	3	2	9
\$2	6	5	4	15

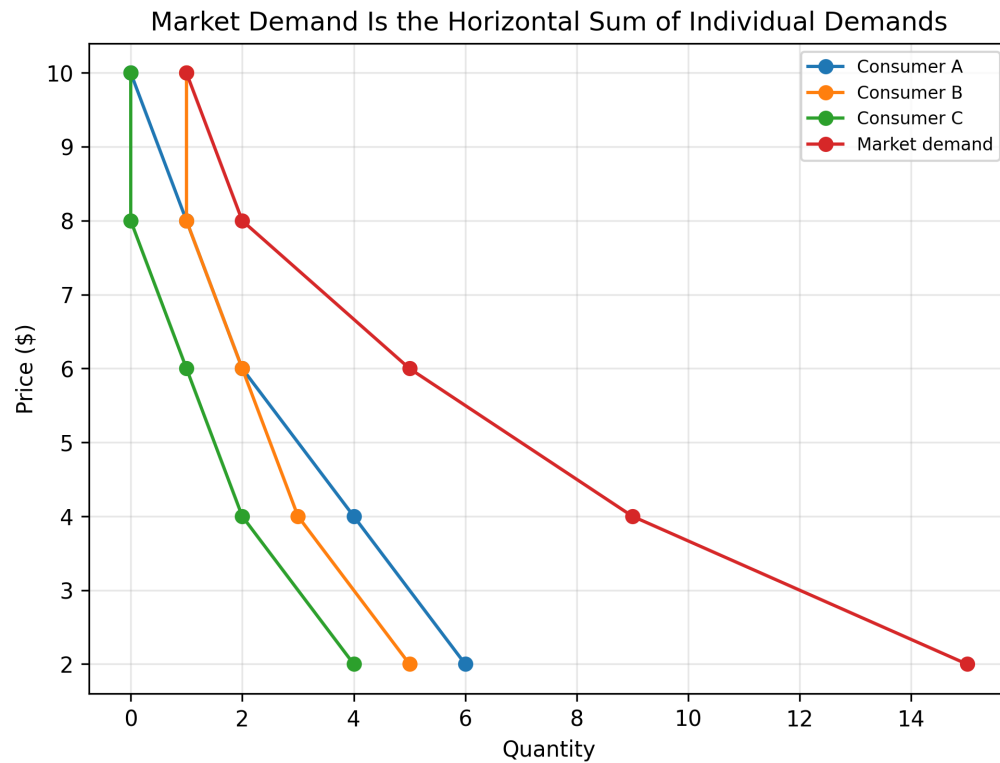


Figure 5: Market demand is the horizontal sum of the quantities demanded by individual consumers.

A market demand curve can shift because existing consumers change their choices or because the number and composition of buyers changes.

9 Movement Along a Demand Curve

When the good's own price changes and all other relevant factors remain constant, the quantity demanded changes. This is represented as a movement along the demand curve.

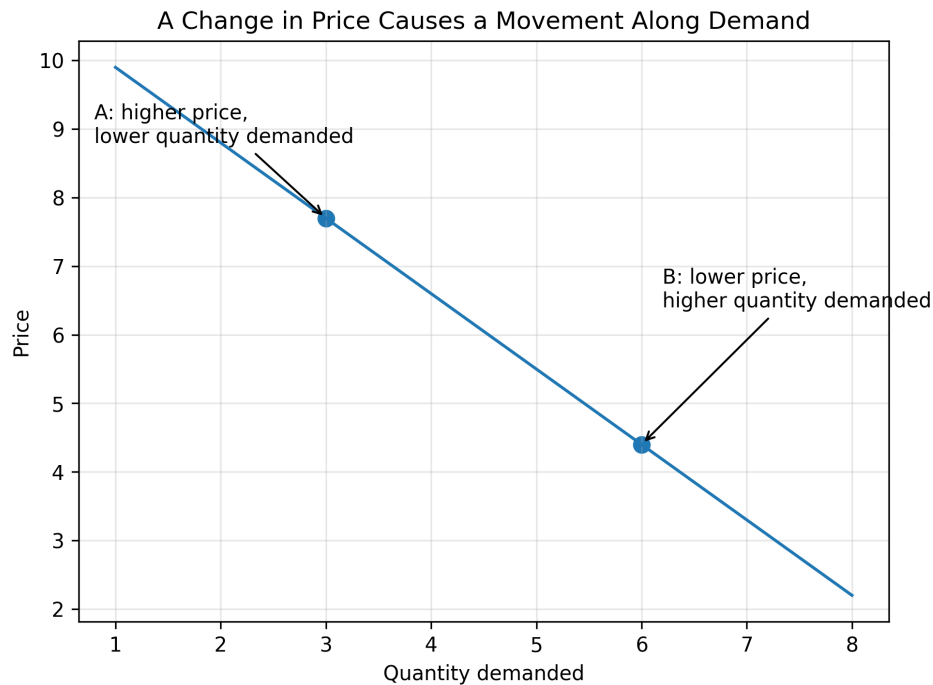


Figure 6: A change in price causes a movement along the demand curve.

For example, if the price of movie tickets falls from \$15 to \$10, consumers may attend more movies. The demand curve itself has not shifted. We are observing a different quantity demanded on the same curve.

10 Shifts in Demand

Demand shifts when a factor other than the good's own price changes the quantity consumers wish to buy at every possible price. An increase in demand shifts the curve to the right. A decrease in demand shifts it to the left.

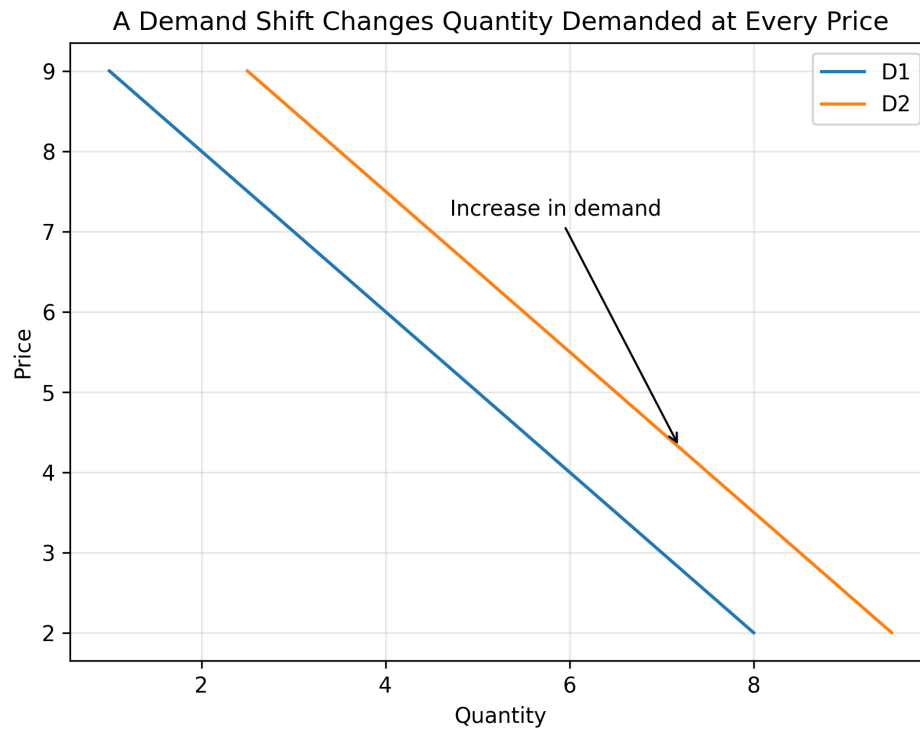


Figure 7: An increase in demand shifts the demand curve to the right.

The major demand shifters include income, tastes and preferences, prices of related goods, expectations, demographics, and the number of buyers.

Major Factors That Shift Demand

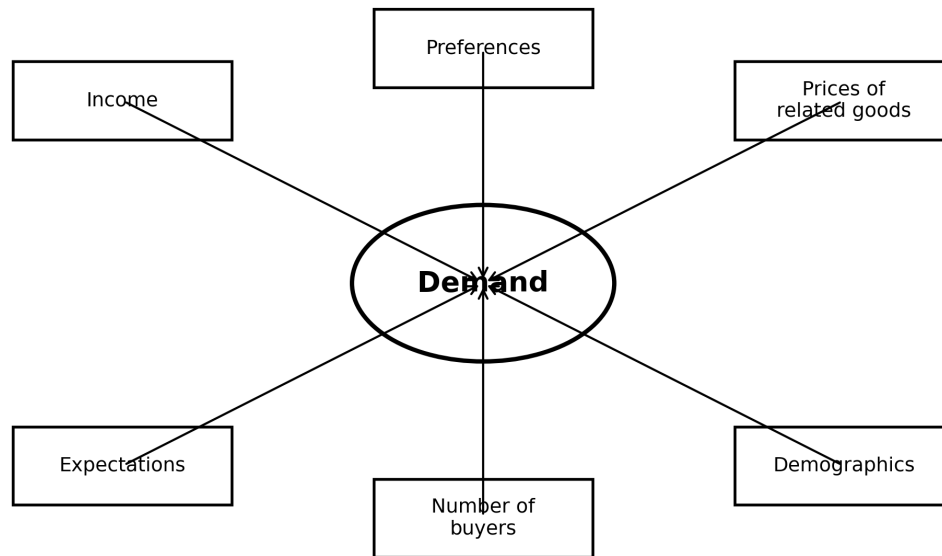


Figure 8: Major factors that can shift demand.

10.1 Income

For a normal good, demand increases when income increases and falls when income falls. Restaurant meals, travel, and many forms of entertainment are often treated as normal goods. For an inferior good, demand may fall as income rises because consumers shift toward alternatives they prefer but previously could not afford. “Inferior” is a technical term and does not mean the good is objectively bad.

10.2 Prices of Related Goods

Two goods are substitutes when a rise in the price of one increases demand for the other. Coffee and tea may be substitutes for some consumers. Two goods are complements when they are often consumed together. A rise in the price of printers may reduce demand for printer ink. A rise in the price of air travel may reduce demand for airport parking.

10.3 Preferences, Expectations, and Buyers

Advertising, information, health concerns, fashion, weather, and social norms can alter preferences. If consumers expect a product’s price to rise next month, they may buy more today. Population growth usually increases market demand, while demographic change may shift demand toward

goods preferred by particular groups.

11 Real-World Connection: Gasoline Demand

Gasoline illustrates both the law of demand and the importance of context. The U.S. Energy Information Administration notes that higher prices generally reduce demand, but gasoline demand may respond only modestly in the short run because many drivers have limited alternatives (U.S. Energy Information Administration, 2014).

A commuter may be unable to move closer to work, replace a vehicle, or change jobs immediately. Over a longer period, consumers can purchase more fuel-efficient vehicles, move closer to work, use public transportation, carpool, or alter travel patterns. This distinction will later lead us to elasticity: how strongly quantity demanded responds to price.

The EIA also explains that unexpectedly lower petroleum consumption in 2020 contributed to lower petroleum-product prices (U.S. Energy Information Administration, 2026). That episode involved both demand and supply, so a complete explanation requires the framework developed across this and the next two lectures.

Real-World Connection

Gasoline shows why the law of demand is not the claim that consumers immediately stop buying when price rises. It says that, other things equal, they buy less than they otherwise would.

12 Real-World Connection: Household Spending

The Consumer Expenditure Survey measures how U.S. households allocate spending across categories such as food, housing, transportation, education, and healthcare (U.S. Bureau of Labor Statistics, 2025).

BLS reported that in 2024 average household spending was \$6,224 on food at home and \$3,945 on food away from home. Average spending was \$225 on sports and exercise equipment and \$272 on gym and health-club memberships (U.S. Bureau of Labor Statistics, 2026).

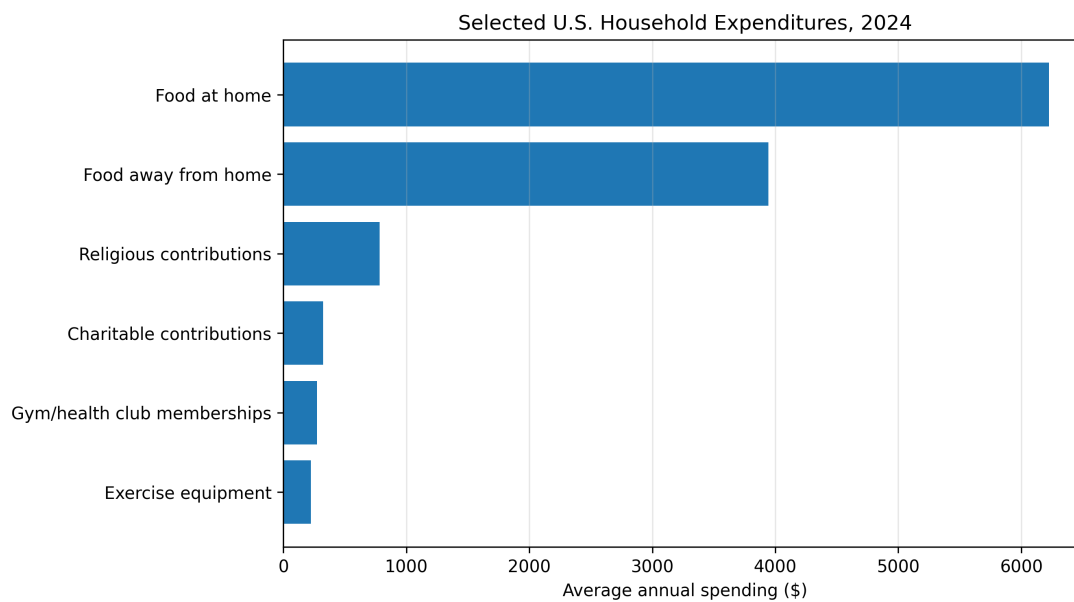


Figure 9: Selected average U.S. household expenditures in 2024. Source: Bureau of Labor Statistics.

These data describe expenditure, not demand curves. Spending equals price multiplied by quantity:

$$\text{Expenditure} = P \times Q.$$

A rise in spending could reflect a higher quantity, a higher price, or both. We should therefore avoid interpreting expenditure changes as quantity changes without additional information.

Common Mistake

Observed spending is not the same thing as quantity demanded. If prices rise sharply, households may spend more while purchasing fewer units.

13 A Practical Method for Reading Demand Stories

When reading a news story about consumer behavior, ask:

1. What is the market?
2. What is the relevant price?
3. What quantity are consumers choosing?
4. Is the story describing a movement along demand or a shift in demand?
5. If demand shifted, which factor changed?

6. Are the data measuring quantities, prices, expenditures, or intentions?
7. What other factors should be held constant?

14 Common Misconceptions

“Demand means desire”

Desire alone is not effective demand. A person may want a beachfront house but be unable to purchase one. Demand requires willingness and ability to pay.

“A higher price changes demand”

A higher own price normally causes a movement along demand and changes quantity demanded. It does not shift the demand curve.

“If sales increased, demand must have increased”

Sales can rise because price fell, supply increased, or both curves shifted. Observing price and quantity together is necessary, and sometimes still insufficient, to identify the cause.

“Every downward-sloping relationship proves the law of demand”

A demand curve is a ceteris paribus relationship. Real-world data often contain simultaneous changes in income, tastes, expectations, and supply. Causal interpretation requires care.

15 Takeaways

1. Demand summarizes the quantities consumers are willing and able to buy at different prices.
2. Quantity demanded refers to one point on that relationship.
3. The law of demand predicts an inverse relationship between price and quantity demanded, holding other factors constant.
4. Willingness to pay helps explain the height of the demand curve.
5. When willingness to pay exceeds the price, consumers receive a benefit from voluntary exchange; we will measure this formally in Lecture 6.
6. Market demand is the horizontal sum of individual demands.

7. A change in the good's own price causes a movement along demand.
8. Changes in income, preferences, related prices, expectations, demographics, or the number of buyers shift demand.
9. Expenditure data combine price and quantity and must be interpreted carefully.

Looking Ahead

Consumers have now answered one question:

How much would I like to buy at different prices?

However, markets also require producers. Who decides how much coffee should be roasted? How many smartphones should be manufactured? How many airline flights should be offered?

The answers come from firms.

In the next lecture we study how producers make decisions and develop the concept of supply. Only after understanding both buyers and sellers will we be able to explain how market prices are determined.

Economic Intuition

What question can we answer now that we could not answer before?

We can now explain why consumers generally buy more when prices fall and identify the factors that shift demand. We still cannot explain where the market price itself comes from. That requires the producer side of the market.

References

- OpenStax. *Principles of Economics 3e*, Section 3.1, “Demand, Supply, and Equilibrium in Markets for Goods and Services.” <https://openstax.org/books/principles-economics-3e/pages/3-1-demand-supply-and-equilibrium-in-markets-for-goods-and-services>
- OpenStax. *Principles of Economics 3e*, Chapter 3 Key Terms and Summary. <https://openstax.org/books/principles-economics-3e/pages/3-key-concepts-and-summary>
- U.S. Bureau of Labor Statistics. “Consumer Expenditure Survey.” <https://www.bls.gov/respondents/cex/>
- U.S. Bureau of Labor Statistics. “The New Year and Household Spending.” 2026. <https://www.bls.gov/opub/ted/2026/the-new-year-and-household-spending.htm>

- U.S. Energy Information Administration. “Gasoline Prices Tend to Have Little Effect on Demand for Car Travel.” 2014. <https://www.eia.gov/todayinenergy/detail.php?id=19191>
- U.S. Energy Information Administration. “What Drives Petroleum Product Prices: Consumption.” <https://www.eia.gov/finance/markets/products/consumption.php>